Vectors I Homework 2: Scalar Product — Solutions

Compiled by Maximus Pung Jun

Official worked solutions for sourced past-paper questions; True/False answers from the Basics Worksheet.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. $a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$	**T**
(b)	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$. $\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=\lvert\mathbf{a}\rvert^2$	**T**
(c)	$\mathbf{i}\cdot\mathbf{j}=0$. $1\cdot0+0\cdot1+0\cdot0=0$	**T**
(d)	$\mathbf{a}\cdot\mathbf{b}$ is a vector. $\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar)	**F**
(e)	$\mathbf{i}\cdot\mathbf{j}=\mathbf{k}$. $\mathbf{i}\cdot\mathbf{j}=0$	**F**
(f)	If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$. $\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$	**T**
(g)	$(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$. $(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\,(\mathbf{a}\cdot\mathbf{b})$	**T**
(h)	$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector. $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$	**T**
(i)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$. Definition of scalar product	**T**
(j)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$. $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar)	**F**
(k)	If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. $\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$	**F**
(l)	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\lvert\mathbf{i}\cdot\mathbf{j}\rvert=0\neq1=\lvert\mathbf{i}\rvert\lvert\mathbf{j}\rvert$	**F**
(m)	$\mathbf{a}\cdot\mathbf{b}\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$	**T**

	Statement	**T/F**
(n)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute. $\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta>0 \therefore \cos\theta>0,\ 0°\le\theta<90°$	**T**
(o)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse. $\cos\theta<0 \therefore 90°<\theta\le 180°$	**T**
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}\ge 0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute. $\mathbf{i}\cdot\mathbf{j}=0,\ \theta=90°$	**F**
(q)	$\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar. $\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless	**F**
(r)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}: 4\ne 2$	**F**
(s)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel. $\cos\theta=1 \therefore \theta=0°$	**T**
(t)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$. $\cos(360°-\theta)=\cos\theta$	**T**
(u)	Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at B	**T**
(v)	The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}: \lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert=1\ne 2=\lvert\mathbf{b}\cdot\hat{\mathbf{a}}\rvert$	**F**
(w)	If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$. $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2=0$	**T**
(x)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. $\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}: \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\ne\mathbf{c}$	**F**
(y)	Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. $\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$	**F**
(z)	Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$. $\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$	**F**

2 [TJC 2025 Prelim P1 Q6]

Official solution.

(a)

Using the ratio theorem, $\overrightarrow{OX}=\dfrac{3\mathbf{a}+\mathbf{b}}{4}$.

(b)

$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\dfrac{\pi}{6}=\lambda\cdot\lambda\cdot\dfrac{\sqrt{3}}{2}=\dfrac{\sqrt{3}}{2}\lambda^2$.

(c)

$$\overrightarrow{OY} = \left(\overrightarrow{OX}\cdot\hat{\mathbf{b}}\right)\hat{\mathbf{b}} = \left(\frac{3\mathbf{a}+\mathbf{b}}{4}\cdot\frac{\mathbf{b}}{|\mathbf{b}|}\right)\frac{\mathbf{b}}{|\mathbf{b}|} = \frac{1}{4|\mathbf{b}|^2}(3\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{b})\,\mathbf{b}$$
$$= \frac{1}{4\lambda^2}\left(3\lambda^2\cos\frac{\pi}{6}+\lambda^2\right)\mathbf{b} = \frac{1}{4}\left(3\cdot\frac{\sqrt{3}}{2}+1\right)\mathbf{b} = \frac{2+3\sqrt{3}}{8}\mathbf{b} \quad \text{(shown)}.$$

(d)

$$\text{Area of } \Delta OXY = \frac{1}{2}\left|\overrightarrow{OX}\times\overrightarrow{OY}\right| = \frac{1}{2}\left|\frac{3\mathbf{a}+\mathbf{b}}{4}\times\frac{2+3\sqrt{3}}{8}\mathbf{b}\right| = \frac{3(2+3\sqrt{3})}{64}|\mathbf{a}\times\mathbf{b}|$$
$$= \frac{3(2+3\sqrt{3})}{64}|\mathbf{a}||\mathbf{b}|\sin\frac{\pi}{6} = \frac{3(2+3\sqrt{3})}{64}\lambda\cdot\lambda\cdot\frac{1}{2} = \frac{3(2+3\sqrt{3})}{128}\lambda^2 \text{ units}^2.$$

3 [HCI 2025 Prelim P2 Q4]

Official solution.

(a)(i)

$\mathbf{c} = -\mathbf{a}-\mathbf{b}$.

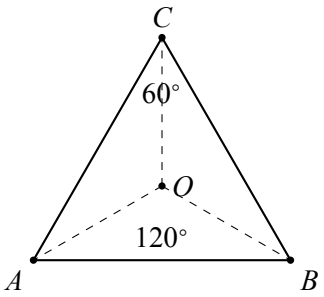
$$\mathbf{c}\cdot\mathbf{c} = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = |\mathbf{a}|^2+2\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2$$
$$1 = 1+2\mathbf{a}\cdot\mathbf{b}+1 \qquad \Rightarrow \qquad \mathbf{a}\cdot\mathbf{b} = -\frac{1}{2}.$$

(a)(ii)

$-\frac{1}{2} = |\mathbf{a}||\mathbf{b}|\cos\angle AOB \Rightarrow \cos\angle AOB = -\frac{1}{2} \Rightarrow \angle AOB = 120°$.

(a)(iii)

Since $\angle AOB = 120°$ and $OA = OB = OC$, the triangles OAB, OAC and OBC are congruent and $\angle ACB = \angle ABC = \angle BCA = 60°$. Hence triangle ABC is an equilateral triangle.



(b)

Since $\overrightarrow{OD} = \mathbf{a}+\mathbf{b} = -\mathbf{c}$ and OD, AB are diagonals of the rhombus $OADB$, $OD \perp AB$. Hence $ACBD$ is a kite.
Official Method 1:

$$\text{Area of } ACBD = 2\cdot\tfrac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{DC}\right| = |(\mathbf{c}-\mathbf{a})\times 2\mathbf{c}| = 2\,|(-\mathbf{b}-\mathbf{a})\times\mathbf{a}|$$
$$= 2\sin 120° = \sqrt{3}.$$

Method 2: Area of $ACBD = \frac{1}{2}\cdot DC\cdot AB = \frac{1}{2}\cdot 2\cdot\sqrt{3} = \sqrt{3}$.

4 [SAJC 2025 Prelim P1 Q11]

Official solution.

(i)

Since AA' is a straight line that passes through O, $\overrightarrow{OA'} = -\mathbf{a}$.

$$\overrightarrow{BA} = \mathbf{a} - \mathbf{b}, \qquad \overrightarrow{BA'} = -\mathbf{a} - \mathbf{b}$$
$$\overrightarrow{BA} \cdot \overrightarrow{BA'} = (\mathbf{a} - \mathbf{b}) \cdot (-\mathbf{a} - \mathbf{b}) = -|\mathbf{a}|^2 - \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{a} + |\mathbf{b}|^2$$
$$= -|\mathbf{a}|^2 + |\mathbf{b}|^2 = 0 \quad (\text{since } |\mathbf{a}| = |\mathbf{b}| = \text{radius}).$$

Hence $\overrightarrow{BA} \perp \overrightarrow{BA'}$, so $\angle ABA' = 90°$ (shown).

(ii)

Area of $\Delta OAB = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$. From the diagram, $\frac{1}{2} \cdot OA \cdot h = \frac{1}{2}|\mathbf{a} \times \mathbf{b}|$.

$$\text{Area of } \Delta ABA' = \frac{1}{2} \cdot AA' \cdot h = \frac{1}{2} \cdot 2\, OA \cdot h = \left(\frac{1}{2} \cdot OA \cdot h\right) \cdot 2 = |\mathbf{a} \times \mathbf{b}|.$$

Thus $k = 1$.

5 [VJC 2025 Prelim P1 Q3]

Official solution.

(a)

$$\text{Area of } \Delta ABD = \frac{1}{2}\left|\overrightarrow{DB} \times \overrightarrow{AB}\right| = \frac{1}{2}\left|\frac{\mu}{\lambda + \mu}\mathbf{b} \times (\mathbf{b} - \mathbf{a})\right| = \frac{\mu}{2(\lambda + \mu)}|\mathbf{a} \times \mathbf{b}|.$$
$$\text{Area of } \Delta ABC = \frac{1}{2}\left|\overrightarrow{CA} \times \overrightarrow{AB}\right| = \frac{1}{2}\left|\frac{1}{3}\mathbf{a} \times (\mathbf{b} - \mathbf{a})\right| = \frac{1}{6}|\mathbf{a} \times \mathbf{b}|.$$

Given area of $\Delta ABD = \frac{1}{2}$ area of ΔABC:

$$\frac{\mu}{2(\lambda + \mu)}|\mathbf{a} \times \mathbf{b}| = \frac{1}{2} \cdot \frac{1}{6}|\mathbf{a} \times \mathbf{b}| \quad \Rightarrow \quad \frac{\mu}{2(\lambda + \mu)} = \frac{1}{12} \quad \Rightarrow \quad \lambda : \mu = 5 : 1.$$

(b)

$$\overrightarrow{AE} = \left(\frac{1}{4}\mathbf{a} + \frac{5}{8}\mathbf{b}\right) - \mathbf{a} = -\frac{3}{4}\mathbf{a} + \frac{5}{8}\mathbf{b},$$
$$\overrightarrow{AD} = \frac{5}{6}\mathbf{b} - \mathbf{a} = \frac{4}{3}\left(-\frac{3}{4}\mathbf{a} + \frac{5}{8}\mathbf{b}\right) = \frac{4}{3}\overrightarrow{AE}.$$

Since $\overrightarrow{AD} = \frac{4}{3}\overrightarrow{AE}$ and A is a common point, A, E and D are collinear.

(c)

Since O lies on the perpendicular bisector of AB, $|\mathbf{a}| = |\mathbf{b}|$.

$$\text{Length of projection of } \mathbf{a} \text{ on } \mathbf{b} = \mathbf{a} \cdot \hat{\mathbf{b}} = |\mathbf{a}| \cos\frac{\pi}{4} = \frac{1}{2}|\mathbf{b}|.$$
$$\overrightarrow{OF} = \frac{1}{2}\mathbf{b}.$$

6 [CJC 2024 Prelim P2 Q1]

Official solution.

(a)

$$\overrightarrow{PQ} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \qquad \overrightarrow{QR} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \\ 1 \end{pmatrix}$$

$$\text{Length of projection of } \overrightarrow{QR} \text{ on } \overrightarrow{PQ} = \frac{|\overrightarrow{QR} \cdot \overrightarrow{PQ}|}{|\overrightarrow{PQ}|} = \frac{\left| \begin{pmatrix} -3 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} \right|}{\sqrt{1+9+4}} = \frac{|-3-9+2|}{\sqrt{14}} = \frac{10}{\sqrt{14}} = \frac{5\sqrt{14}}{7}.$$

(b)

$$|\mathbf{a}+\lambda\mathbf{b}|^2 = 4|\mathbf{a}-\lambda\mathbf{b}|^2$$
$$(\mathbf{a}+\lambda\mathbf{b})\cdot(\mathbf{a}+\lambda\mathbf{b}) = 4(\mathbf{a}-\lambda\mathbf{b})\cdot(\mathbf{a}-\lambda\mathbf{b})$$
$$|\mathbf{a}|^2+2\lambda\mathbf{a}\cdot\mathbf{b}+\lambda^2|\mathbf{b}|^2 = 4(|\mathbf{a}|^2-2\lambda\mathbf{a}\cdot\mathbf{b}+\lambda^2|\mathbf{b}|^2)$$

$$|\mathbf{a}|^2+2\lambda\mathbf{a}\cdot\mathbf{b}+\lambda^2|\mathbf{b}|^2 = 4|\mathbf{a}|^2-8\lambda\mathbf{a}\cdot\mathbf{b}+4\lambda^2|\mathbf{b}|^2$$
$$10\lambda\mathbf{a}\cdot\mathbf{b} = 3|\mathbf{a}|^2+3\lambda^2|\mathbf{b}|^2$$
$$10\lambda\,|\mathbf{a}||\mathbf{b}|\cos\theta = 3\lambda^2|\mathbf{b}|^2+3\lambda^2|\mathbf{b}|^2 \qquad (|\mathbf{a}| = \lambda|\mathbf{b}|)$$
$$10\lambda\cdot(\lambda|\mathbf{b}|)\,|\mathbf{b}|\cos\theta = 6\lambda^2|\mathbf{b}|^2$$
$$\cos\theta = \frac{3}{5} \quad \text{(shown).}$$

END OF SOLUTIONS